Positive inputs suffice for induced Schatten quasi-norms

A full answer to an open question in arXiv:2604.14055

Automated mathematics discovery run `fa_banach_001`
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Status: candidate full solution, likely valid. For every completely positive map between finite-dimensional matrix algebras and every pair of positive Schatten exponents, the induced quasi-norm equals its restriction to positive semidefinite inputs. This fully answers the ordinary $q \to p$ question posed in the source, but not its separate completely bounded mixed-quasi-norm counterpart.

1 Source question

Kochanowski, Fawzi and Rouzé introduce two-indexed Schatten quasi-norms and their completely bounded variants [1]. In their Summary and Outlook, on page 49, they ask whether a completely positive (CP) map satisfies

$$\|\Phi\|_{q\to p} = \|\Phi\|^{+}_{q\to p} \qquad (0 < p, q < 1),$$

where the superscript $+$ restricts the input to positive semidefinite operators. They note that the equality was known in the norm range $p, q \geq 1$ [2].

> In Section 4.4.1 we discussed some open questions regarding the achievability of quasi-norms for CP maps on positive operators. In our notations this is effectively asking whether
>
> $$\|\Phi\|_{q\to p} = \|\Phi\|^{+}_{q\to p}$$
>
> is true also for indices $1 > q, p > 0$. For $1 \leq q, p \leq \infty$ see [38] and references therein. Likewise one may ask if the same holds for the completely bounded counterparts we introduced. There see [12, Theorem 12, 13] and [14, Theorem 4.1] for the cases $1 \leq q, p \leq \infty$.

We prove the equality for all positive exponents. The argument does not use duality or the triangle inequality, and so crosses the quasi-norm boundary without modification.

2 Statement

For $0 < r < \infty$, write

$$\|X\|_r = \left(\operatorname{Tr}|X|^r\right)^{1/r}, \qquad \|X\|_\infty = \|X\|_{\mathrm{op}}.$$

For a linear map $\Phi : M_d(\mathbb{C}) \to M_m(\mathbb{C})$, set

$$\|\Phi\|_{q\to p} = \sup_{X\neq 0} \frac{\|\Phi(X)\|_p}{\|X\|_q},$$
$$\|\Phi\|^+_{q\to p} = \sup_{0\neq A\geq 0} \frac{\|\Phi(A)\|_p}{\|A\|_q}.$$

Theorem 1 (Positive-input principle)**.** *Let $\Phi : M_d(\mathbb{C}) \to M_m(\mathbb{C})$ be completely positive. For every $0 < p, q \leq \infty$,*

$$\boxed{\|\Phi\|_{q\to p} = \|\Phi\|^+_{q\to p}\,.}$$

Both suprema are maxima. In fact, complete positivity can be weakened to 2-positivity.

Thus Theorem 1 includes, and fully answers, the source's open range $0 < p, q < 1$.

3 Proof intuition

The polar decomposition of an arbitrary input X places it in the off-diagonal corner of a positive 2×2 block matrix whose diagonal corners are $|X^*|$ and $|X|$. Two-positivity sends that matrix to another positive block matrix. Positivity forces its off-diagonal corner $\Phi(X)$ to factor through the geometric mean of the two positive diagonal corners. Generalized Schatten Hölder then bounds $\|\Phi(X)\|_p$ by the geometric mean of the output quasi-norms at the two positive inputs $|X^*|$ and $|X|$. Those inputs have exactly the same Schatten q-quasi-norm as X, so one of the two positive-input ratios is at least as good as the ratio at X.

The key point is that generalized Schatten Hölder is valid for every positive exponent, including $p < 1$; it is Lemma 2.1 of the source [1].

4 Two block-matrix lemmas

Lemma 2 (Canonical positive block)**.** *For every $X \in M_d(\mathbb{C})$,*

$$\begin{pmatrix} |X^*| & X \\ X^* & |X| \end{pmatrix} \geq 0.$$

Proof. Let $X = U|X|$ be the polar decomposition. Since $|X^*| = U|X|U^*$, one has the Gram factorization

$$\begin{pmatrix} |X^*| & X \\ X^* & |X| \end{pmatrix} = \begin{pmatrix} U \\ I \end{pmatrix} |X| \begin{pmatrix} U^* & I \end{pmatrix},$$

which is positive semidefinite. □

Lemma 3 (Off-diagonal estimate)**.** *If*

$$\begin{pmatrix} A & B \\ B^* & D \end{pmatrix} \geq 0, \qquad A, D \geq 0,$$

then there is a contraction K such that

$$B = A^{1/2} K D^{1/2}.$$

Consequently, for every $0 < p \leq \infty$,

$$\|B\|_p \leq \|A\|_p^{1/2} \|D\|_p^{1/2}\,. \tag{1}$$

Proof. For $\varepsilon > 0$, the block matrix remains positive after adding εI to both diagonal corners. The Schur complement therefore shows that

$$K_\varepsilon = (A + \varepsilon I)^{-1/2} B (D + \varepsilon I)^{-1/2}$$

is a contraction. In finite dimensions, a subsequence converges to a contraction K. Passing to the limit in

$$B = (A + \varepsilon I)^{1/2} K_\varepsilon (D + \varepsilon I)^{1/2}$$

gives $B = A^{1/2} K D^{1/2}$, including when A or D is singular.

Generalized Schatten Hölder, first for the product of $A^{1/2}$ and $KD^{1/2}$ and then with the operator norm of K, gives

$$\begin{aligned} \|B\|_p &\leq \left\|A^{1/2}\right\|_{2p} \left\|KD^{1/2}\right\|_{2p} \\ &\leq \left\|A^{1/2}\right\|_{2p} \|K\|_\infty \left\|D^{1/2}\right\|_{2p} \\ &\leq \|A\|_p^{1/2} \|D\|_p^{1/2} . \end{aligned}$$

Here $2p = \infty$ when $p = \infty$. Generalized Hölder holds for $0 < p < 1$ as well as for the norm range, so no convexity is being used. $\square$

5 Proof of Theorem 1

Fix $X \in M_d(\mathbb{C})$. Applying the 2-positive map $\mathrm{id}_{M_2} \otimes \Phi$ to the positive matrix in Lemma 2 yields

$$\begin{pmatrix} \Phi(|X^*|) & \Phi(X) \\ \Phi(X)^* & \Phi(|X|) \end{pmatrix} \geq 0.$$

We used here that every positive linear map is adjoint preserving. By Lemma 3,

$$\|\Phi(X)\|_p \leq \|\Phi(|X^*|)\|_p^{1/2} \|\Phi(|X|)\|_p^{1/2} . \tag{2}$$

The three matrices X, $|X|$, and $|X^*|$ have the same singular values. Hence

$$\|X\|_q = \||X|\|_q = \||X^*|\|_q \qquad (0 < q \leq \infty).$$

Dividing (2) by this common value gives

$$\begin{aligned} \frac{\|\Phi(X)\|_p}{\|X\|_q} &\leq \left(\frac{\|\Phi(|X^*|)\|_p}{\||X^*|\|_q}\right)^{1/2} \left(\frac{\|\Phi(|X|)\|_p}{\||X|\|_q}\right)^{1/2} \\ &\leq \|\Phi\|_{q\to p}^+ . \end{aligned}$$

Taking the supremum over $X \neq 0$ proves $\|\Phi\|_{q\to p} \leq \|\Phi\|_{q\to p}^+$. The reverse inequality is immediate because positive inputs are a subset of all inputs.

Finally, in finite dimensions the unit Schatten q-sphere and its positive part are compact, while the objective is continuous. Thus the suprema are attained. Only 2-positivity was used.

6 Scope and novelty check

This is a full answer to the ordinary induced Schatten $q \to p$ question in the Summary and Outlook of [1], and it strengthens the requested range from $0 < p, q < 1$ to all $0 < p, q \leq \infty$. It does *not* settle the separate question about the source's completely bounded mixed two-indexed quasi-norms. It also does not address the interpolation conjecture on the same source page or infinite-dimensional extensions.

The run's cheap indexes were searched by arXiv identifier, exact title, and the core phrases "completely positive map", "Schatten quasi-norm", and "positive input". A bounded primary-source search located the source and the norm-range result [2], but no paper recording the $p, q < 1$ extension or this two-by-two proof. This is not an exhaustive novelty certification.

References

[1] J. Kochanowski, O. Fawzi and C. Rouzé, *Two-Indexed Schatten Quasi-Norms with Applications to Quantum Information Theory*, arXiv:2604.14055 (2026).

[2] J. Watrous, *Notes on super-operator norms induced by Schatten norms*, arXiv:quant-ph/0411077 (2004).